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Review

Early human embryonic development: Blastocyst formation to gastrulation

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SUMMARY

There has been recent renewed interest in studying human early embryonic development. The advent of improved culture conditions to maintain blastocysts *in vitro* for an extended period and the emerging stem-cell-based models of the blastocyst and peri-implantation embryos have provided new information that is relevant to early human embryogenesis. However, the mechanism of lineage development and embryonic patterning, and the molecular pathways involved in their regulation, are still not well understood. Interest in human embryonic development has been reinvigorated recently given numerous technical advances. In this review, Rossant and Tam discuss new insights into human embryogenesis gathered from successes in culturing human embryos *in vitro* and stem-cell-based embryo models. Then they outline what questions still need answering.

INTRODUCTION

During the first 5 to 6 days after fertilization and prior to implantation, the human zygote develops into a blastocyst that comprises an outer layer of trophectoderm (TE) enclosing the blastocyst cavity in which a population of inner cell mass (ICM) cells is juxtaposed asymmetrically to the TE on one side of the blastocyst. The ICM contains two types of cells, the epiblast and the hypoblast (generally known as primitive endoderm in rodents), with the latter forming an epithelial layer on the luminal side of the epiblast. By analogy to the mouse (Cockburn and Rossant, 2010), the TE is predicted to give rise to the trophoblast cells in the placenta, and the epiblast contributes to cells and tissues of the embryo proper and, together with the hypoblast, the extraembryonic membranes of the conceptus.

Further development of the blastocyst necessitates the establishment of connectivity with the maternal uterine environment for the provision of embryotrophic support. This is accomplished by the implantation of the blastocyst via trophoblast-uterine tissue interaction and embedding of the blastocyst in the uterine stroma. This peri-implantation period is followed by the events of gastrulation, placentation, and postgastrulation organogenesis: critical phases of embryonic development during which the blueprint of development (the body plan) is established through the ordered generation of diverse cell types and the assembly of the precursor tissues for body parts and organs.

Following implantation, when the blastocyst is embedded in the uterine tissue, the embryo becomes inaccessible for experimental study. Much of the knowledge on early human embryogenesis was gleaned from descriptive study of the morphology of rare embryos collected during the period of peri-implantation development to gastrulation (Carnegie stage CS5–7 embryos) (O'Rahilly and Muller, 1987). Further research on the peri-implantation development of human embryos has been beset by the paucity of materials and technical feasibility of procuring specimens. Advances in the methodology for the *in vitro* culture of developing blastocysts over the implantation period (Deglincerti et al., 2016; Shahbazi et al., 2016; Xiang et al., 2020) have opened up new possibilities for studying this important time window (Figure 1A). However, research on human embryonic material is subject to various ethical and regulatory constraints in different jurisdictions that may place limits on both the access to materials and the extent of experimental analysis *in vitro* that is allowed. In particular, the so-called 14-day rule, which limits the culture of intact human embryos to 14 days postfertilization (dpf) and/or the initiation of gastrulation, has been widely applied and is enshrined in legislation in many jurisdictions (Matthews and Moralí, 2020). It was also included as a recommended ban in the international guidelines produced by the International Society for Stem Cell Research in 2016. However, the recent revision of the ISSCR Stem Cell Guidelines (Lovell-Badge et al., 2021) has suggested that, if there is a clear and convincing scientific rationale for the need for an extended culture period, a blanket ban could be replaced with a broad consultative review process, subject to local laws and regulations (Clark et al., 2021). The recent development of stem-cell-based integrated and nonintegrated embryo models (Rossant and Tam, 2021; Weatherbee et al., 2021) provides new avenues for human embryology research with potentially fewer ethical concerns but with a clear need to be validated against normal embryonic processes. Scientific knowledge of the cellular and molecular mechanism of embryogenesis will inform the mechanistic basis of errors of development in genetic disorders and nongenetic congenital

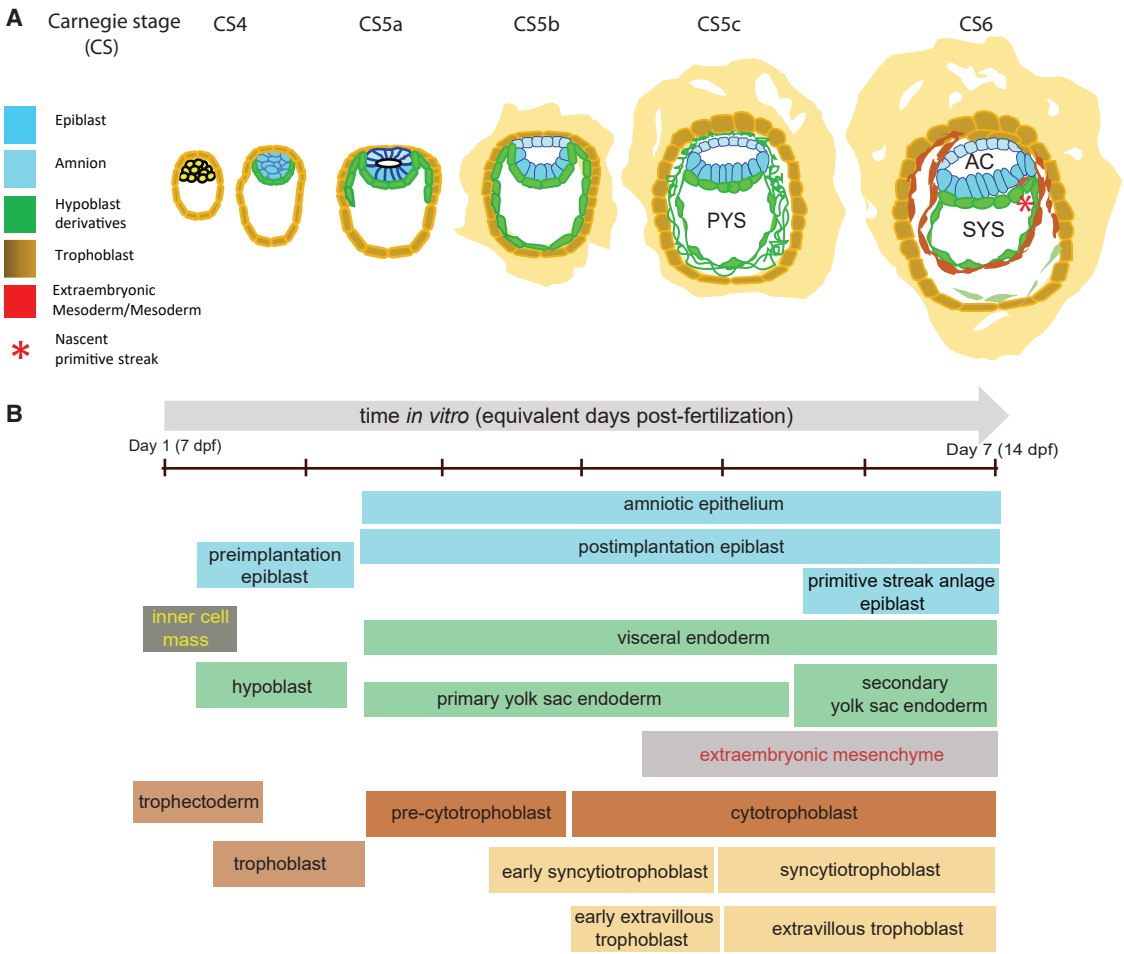


Figure 1. Developmental milestones and diversification of cell/tissue types in peri-implantation human embryos
(A) Morphogenesis of the human embryo developing from the blastocyst stage to Stage CS7 embryo featuring the tissue compartments of the gastrula-stage embryo. CS, Carnegie stage; AC, amniotic cavity; PYS, primary yolk sac; SYS, secondary yolk sac; Emb, embryonic disc. Color legend: cell and tissue types. (B) Cells and tissues in the peri-implantation human embryo *in vitro*. The approximate time window when different types of cells and tissues are present in the cultured embryo is indicated by the length of the bar.

diseases, as well as shedding light on the causes of early embryo loss and pregnancy disorders. Here, we provide a snapshot of the recent learnings of early human embryogenesis from blastocyst formation and across the immediate post-implantation period of development.

MAKING THE BLASTOCYST: THE PREIMPLANTATION LINEAGES

The preimplantation stages of human development have been studied for many years (Rossant and Tam, 2017), following the development of *in vitro* culture conditions for fertilized embryos for human assisted reproduction (Johnson, 2019). The timing of the onset of zygotic genome activation, cleavage-stage cell divisions, the morphological changes of compaction and blastocoel formation, and the visible delineation of the blastocyst cell lineages have all been documented (Gerri et al., 2020b; Shahbazi, 2020). Zygotic genome activation occurs around the 4–8 cell stage in the human embryo (Braude et al., 1988), versus the 2-cell stage in the mouse (Schultz, 1993), and succeeding events of compaction, polarization, and blastocoel formation are also delayed compared with the mouse. Compaction and establishment of apical polarity marks the initial separation of outer polarized cells that will go on to form the TE and takes place between days 3 and 4 after fertilization in the human (Iwata et al., 2014). A recent study has detailed the dynamics of compaction in the human embryo at the 8–16 cell stage and showed that transmembrane phosphoinositide-specific phospholipase C (PLC) signaling is required for apical domain formation (Zhu et al., 2021). In final form and function, the human blastocyst at embryonic day (E) 6–7, with its three distinct cell lineages, is very similar to the mouse blastocyst, but the detailed mechanisms of lineage commitment may differ. The knowledge of the events of lineage specification in the human embryo is still quite rudimentary compared with that in the mouse. Although some of the major genes involved in lineage specification in the mouse, such as *Pou5f1 (Oct4), Nanog, Cdx2*, and *Gata6*, are conserved and expressed in the early human embryo (Blakeley et al., 2015; Niakan and Eggan, 2013), the timing of their expression in relation

to lineage specification is not always the same as in the mouse. *OCT4*, which is a key transcription factor (TF) for ICM formation in mice (Nichols et al., 1998), is broadly expressed in all cells, including both ICM and TE, right up to the expanded blastocyst stage, while *CDX2*, which is a key regulator of TE initiation in the mouse (Strumpf et al., 2005), is not expressed until after blastocyst formation (Niakan and Eggan, 2013). Other genes involved in TE formation in the mouse, such as *Elf5* and *Eomes*, are not expressed at the blastocyst stage in humans, while *GATA3*, which plays a subsidiary role in mouse TE formation (Ralston et al., 2010), is expressed in TE precursors in the human embryo (Gerri et al., 2020a; Petropoulos et al., 2016; Zhu et al., 2021).

Examination of the expression of individual genes by immunofluorescence in human embryos *in vitro* provided the first indications of differences in expression profiles during blastocyst formation between humans and mice (Niakan and Eggan, 2013). This has been confirmed and expanded with increasingly more detailed information on the patterns and trajectories of gene expression from single-cell RNA-seq analysis of human embryos (Blakeley et al., 2015; Petropoulos et al., 2016; Stirparo et al., 2018; Yan et al., 2013). A recent compilation of published and new data, mapped against morphological landmarks of blastocyst development, has produced a pseudotime trajectory that indicates a separation of ICM and TE lineages only after the blastocyst has formed and begun to expand (Meistermann et al., 2021). This is in contrast to the mouse where a similar analysis shows initial separation of ICM and TE gene expression profiles at the late morula stage (Posfai et al., 2017). The timing of separation of the epiblast and hypoblast lineages in humans is less clear. An earlier scRNA-seq study had suggested that the epiblast/hypoblast segregation event in the human embryo actually coincided with the separation of the ICM and TE lineages (Petropoulos et al., 2016). However, the recent study seems to show a brief, common ICM-progenitor window before divergence of the epiblast and the hypoblast (Meistermann et al., 2021).

This delay in the consolidation of the epiblast, hypoblast, and TE gene expression profiles might suggest that the stabilization of lineage fate could also be delayed compared with the mouse (Rossant and Tam, 2017). There are still very limited experimental data on testing lineage commitment in human embryos, but some studies are consistent with lineage plasticity at the early blastocyst stages. Outside cells isolated from E5 human blastocysts can still regenerate a blastocyst with ICM cells (De Paepe et al., 2013) and isolated E6 ICMs can generate trophoblast outgrowths in culture (Guo et al., 2021). Given that the segregation of ICM and TE in the mouse embryo is driven by the changes in cell polarity and cell adhesion, leading to the separation of inside and outside cell populations during late cleavage stages (reviewed Sasaki 2017), the entire process of ICM/TE segregation in the human embryo could be distinct from that in mice. However, a recent study has implicated the HIPPO-YAP-αPKC pathway in driving the formation of the outer cell layer of the late morula/early blastocyst in humans, in a similar manner to the mouse (Gerri et al., 2020a). At this stage, the scRNA-seq studies would indicate that ICM and TE transcriptional profiles are not distinct, suggesting that these early events are post-transcriptional, driving the morphological events of blastocyst formation before ICM and TE lineage fate is specified.

Lineage specification is usually associated with the expression and function of lineage-specific TFs. However, there is still uncertainty as to the key TFs driving cell fate in the human blastocyst. Only one TF (OCT4) has been mutated so far in developing human embryos. CRISPR-driven knockout of *OCT4* demonstrated that it was required for normal blastocyst development but that the requirement seemed to be earlier than would be predicted if its only role was to specify ICM fate (Fogarty et al., 2017). Expression of *CDX2* in humans was dependent on the presence of *OCT4*, whereas in mice, *Cdx2* and *Oct4* show reciprocal repression at the blastocyst stage (Niwa et al., 2005; Strumpf et al., 2005). GATA3 may replace the role of CDX2 in the initial phases of TE specification, since it did show some early restriction to outside cells and its expression was reduced when αPKC was inhibited at the morula stage (Gerri et al., 2020a; Zhu et al., 2021). *GATA6* expression in the developing hypoblast (Boroviak et al., 2018; Petropoulos et al., 2016; Xiang et al., 2020) suggests a conserved role with the mouse (Koutsourakis et al., 1999; Morrisey et al., 1998), but there are yet no experimental data to back this up in the human embryo. In the mouse the separation of epiblast and primitive endoderm is dependent on local levels of FGF/ERK signaling received by cells in the ICM (Saiz et al., 2016; Yamanaka et al., 2010). However, at least two studies have been unable to demonstrate any dependence on FGF/ERK signaling for establishing cell fate in the human ICM (Kuijk et al., 2012; Roode et al., 2012). What signals replace FGF/ERK in the human ICM is not known.

There are still many gaps in our knowledge of the molecular processes of blastocyst formation in humans that might be augmented by more experimental study. However, there are many restrictions worldwide on access and use of human embryos for research that can limit discovery. Other mammalian species, particularly nonhuman primates, have been studied, but access to and costs of such studies can be prohibitive and differences from humans still exist. Hence, the excitement in reports of generating stem-cell-derived embryo models of the blastocyst stage of development: so-called blastoids. Formation of structures resembling mouse blastocysts from aggregates of mouse embryonic stem (ES) cells and trophoblast stem (TS) cells were first reported by the Rivron lab (Rivron et al., 2018). Primitive endoderm was underrepresented and although these structures could implant in the uterus, they did not show further development. Blastoids with improved inclusion of primitive endoderm were produced when mouse extended-potential ES cells, which are claimed to possess extraembryonic potential (Yang et al., 2017a, 2017b), were either combined with TS cells (Sozen et al., 2019) or cultured in suspension culture alone (Li et al., 2019). These blastoids were morphologically more like a blastocyst but still failed to develop far after implantation, and in the case of the entirely ES-derived structures, gene-expression analysis did not support a bona fide TE identity of all of the outside cells (Posfai et al., 2021b).

Clearly, even in the mouse, the blastoid is not yet equivalent to the blastocyst. Can human blastoids be better models? The logical place to start with generation of human blastoids might be expected to be combinations of human blastocyst-derived stem cells, in a similar manner to the mouse. Human TS cells

have been generated from blastocysts and early chorionic villus biopsies under different culture conditions than those used to generate mouse TS cells (Okae et al., 2018). While they are useful models to study trophoblast differentiation, they are considered to be more related to post-implantation cytotrophoblasts than the TE of the blastocyst (Castel et al., 2020; Mischler et al., 2021; Okae et al., 2018). Perhaps because of this, there have been no published reports on the possible generation of human blastoids from combining human ES and TS cells. Sozen et al. attempted to repeat their success in promoting full lineage development in mouse blastoids with combination of extended potential ES cells with TS cells (Sozen et al., 2019) by making similar aggregates with human cells (Sozen et al., 2021). However, they report poor development and maintenance of the outer TE cells in such aggregates. Instead, there have been a series of reports claiming to generate blastoids directly from human ES cells by manipulating culture conditions but without addition of any specific extraembryonic cell types (Fan et al., 2021; Liu et al., 2021; Kagawa et al, 2021; Yanagida et al., 2021; Yu et al., 2021). One study claimed to produce iblastoids as an intermediate during the process of reprogramming adult cells to induced pluripotent stem cells (iPSCs) (Liu et al., 2021), while most other studies began with human ES cells in a putative naive state. Several reports had previously suggested that ES cells in the naive state retained some potential to differentiate down the TE pathway, depending on appropriate culture conditions (Cinkornpumin et al., 2020; Dong et al., 2020; Guo et al., 2021; Io et al., 2021). This plasticity of the human naive state is not shown by mouse naive ES cells and might relate to the demonstrated plasticity of the ICM in the intact human blastocyst. Some of the reported human blastoids do show quite a striking resemblance to the blastocyst itself in both morphology and gene expression, but in-depth comparisons of the gene-expression profile of the cell types in the blastoid with the embryo itself always need to be made. Formation of an epithelial cyst with enclosed pluripotent cells is not sufficient to confirm a functional TE phenotype. The post-implantation development of human blastoids cannot be tested *in vivo*, but 3D *in vitro* outgrowths should provide a good comparison with intact embryos (Kagawa et al, 2021). Much more data and careful comparison with normal blastocysts and early post-implantation stages are needed to validate these potentially powerful experimental models.

The ability to avoid the use of human embryo material and to scale up production to a level suitable for drug testing and other studies makes the pursuit of the blastoid a likely hot topic in the coming years. However, although regulatory and ethical concerns with the use of human embryos in research could be avoided by use of stem-cell-derived blastoids, they do come with their own legal and ethical concerns. Although blastoids, even in the mouse, are currently far from being functional embryo equivalents, there is some concern about defining when a stem cell model has crossed the line and become an embryo. In some jurisdictions, blastoids would require the same regulatory oversight as embryos themselves (Matthews and Moralí, 2020). The recently revised ISSCR Stem Cell Guidelines took these issues into consideration and proposed that so-called integrated stem cell models like blastoids (with cell types potentially able to generate a functional placenta and embryo) should be subject to special oversight and restricted to short-term culture (Lovell-Badge et al., 2021). It is also specifically prohibited to transfer any stem-cell-based embryo model to the uterus of a human or animal host.

TRACKING DEVELOPMENTAL TRAJECTORIES OF EMBRYONIC AND EXTRAEMBRYONIC CELL TYPES FROM PERI-IMPLANTATION TO GASTRULATION

The refinement of embryo culture methods that support *in vitro* development of peri-implantation human embryos opens an avenue for directly exploring cell-lineage relationships in the human embryo over the immediate post-implantation period. Notwithstanding that morphogenesis of the cultured embryos did not fully recapitulate that of their *in vivo* counterparts, the developmental anatomy largely corroborates that documented for archived specimens (https://www.ehd.org/developmental-stages/stage6.php). The blastocysts can, without the support of uterine tissues, remodel the epiblast into an amniotic sac and form the yolk sac, and the TE differentiates into multiple trophoblast cell types associated with the maternal-fetal interface (Deglincerti et al., 2016; Shahbazi et al., 2016). A more recent study of human blastocysts developed in a 3D culture system has captured a better modeling of the morphogenesis of the peri-implantation embryo up the equivalent 14 dpf (Xiang et al., 2020) (Figure 1A). Sequencing analysis of the transcriptome of 555 cells from 42 3D cultured embryos has identified several major cell types in embryos (Figure 1B). A comparison with the range of cell types found in 6–17 pdf *Cynomolgus* monkey embryos shows that equivalent counterparts of the human embryonic cell types are present in the *Cynomolgus* embryo (Ma et al., 2019; Nakamura et al., 2016; Niu et al., 2019). However, some cell types in the *Cynomolgus* embryo, such as the post-implantation late epiblast, gastrulating 1, -2a, and -2b cells, appeared not have been identified in the cultured human embryo, and some annotated cell types of the cultured human embryo appeared to be different from their counterparts in the monkey embryo (Chhabra and Warmflash, 2021). It is not clear whether this discrepancy is related to the depth of cell coverage and/or the innate species-specific difference in cell-lineage specification and differentiation.

Embedded in the single-cell transcriptome data is an indication of the lineage descendance among cell populations. Based on information regarding the proximity/connectivity of transcriptome features, it is feasible to reconstruct the developmental trajectory of cell populations in the embryo during peri-implantation to gastrulation (Figure 2). However, the timeline of *in vitro* embryonic development is restricted to within the equivalent of the 7–14 dpf period, and therefore the dataset may not cover the cell types that are present in more advanced embryos at gastrulation. A recent study of an intact gastrula-stage embryo (16–19 dpf, Carnegie stage CS7) has generated a spatially resolved transcriptome of cells of the epiblast, ectoderm, mesoderm, endoderm, neural plate, posterior, hypoblast, yolk sac, and primitive streak (PS) and specific cell types such as primordial germ cells (PGCs), erythrocytes, and hemogenic endothelial progenitors (Tyser et al., 2021). This dataset thus provides a meaningful developmental end point of the lineage trajectories of cells of the peri-implantation embryo. By analyzing the connectivity of the single-cell transcriptome of the cultured human embryos and the CS7 embryo, it might be possible to infer the antecedent-

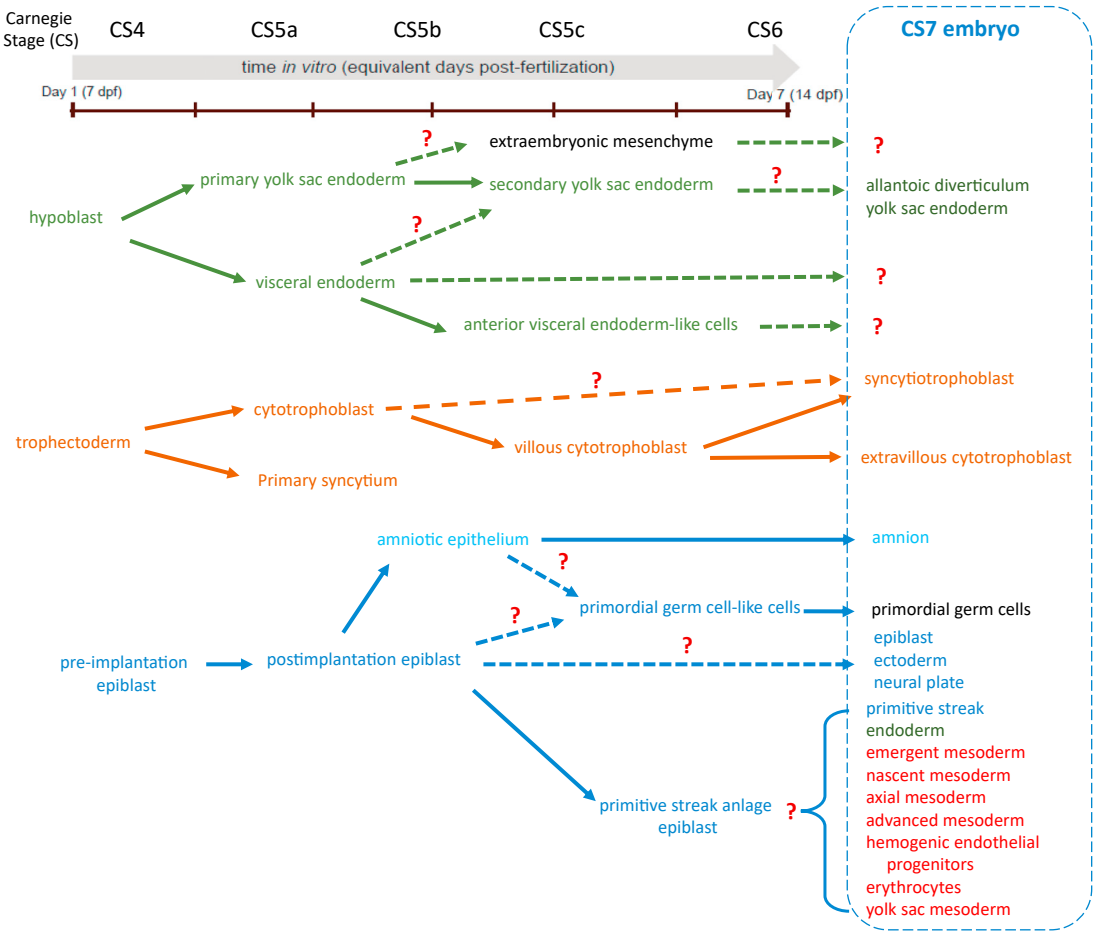


Figure 2. Lineage phylogeny of the human peri-implantation embryo

Predicted lineage relationship of the derivatives of preimplantation epiblast, hypoblast, and trophectoderm in the peri-implantation period based on the anatomical observations of archived specimens and the molecular phenotype of cells revealed by the transcriptome profile. The Carnegie stage (CS) and the approximate *in vitro* timeline when different types of cells and tissues are present are indicated. Uncertain lineage relationships (dashed arrow and "?") between cell and tissue types in the peri-implantation embryo and the CS7 gastrula-stage embryo (dashed box) are highlighted.

descendant relationship of the embryonic and extraembryonic cell populations in the peri-implantation embryos and the gastrula-stage embryo (Figure 2). Integrating this combined dataset with that of preimplantation embryos (Meistermann et al., 2021), a comprehensive compendium of the developmental trajectories of cell populations from the zygote to the gastrula-stage embryo may soon be constructed.

Hypoblast to primary and secondary yolk sac

While the timing and signaling pathways involved in establishing the hypoblast lineage at the blastocyst stage in humans are still to be fully resolved, our understanding of the next phases of hypoblast development into the primary and secondary yolk-sac lineages post-implantation is even more shrouded in mystery and controversy (Ross and Boroviak, 2020). Given that hypoblast derivatives may provide key signals to pattern the epiblast at gastrulation and that the secondary yolk sac (SYS) is the site of early hematopoiesis, this is clearly a subject in need of more exploration.

Much of the interpretation of the events of human early yolk-sac development is based on studies of sectioned material from human embryos in the Carnegie collection (O'Rahilly and Muller, 1987) and other rare sources (Rock and Hertig, 1948) and comparison with sections of early post-implantation monkey embryos (Enders and King, 1988; Enders et al., 1986). As the blastocyst expands and begins the process of attachment and implantation in the uterus, the hypoblast spreads on the inside of the TE to form the bilayered primary yolk sac (Figure 1A). The migrating hypoblast cells, the parietal endoderm, are elongated and loosely attached to each other and the underlying trophoblast. This primary yolk-sac structure is superficially similar to the parietal yolk sac in rodents. In the more advanced human primary yolk sac, a meshwork of cells and cellular processes emerges between the parietal endoderm and the trophoblast. Whether these cells should be considered to be extraembryonic mesoderm, as has been reported in some studies, is unclear. This cellular meshwork is not seen in comparable stages of rhesus development (Enders et al., 1986; Luckett, 1978).

By the CS6 stage, the hypoblast underneath the epiblast of the developing blastodisc forms a compact epithelial layer, the visceral endoderm, which lines part of the new cavity, the secondary yolk sac (SYS) (Figure 1A). Exactly how the SYS forms

is unclear; histology suggests a possible budding off from the primary yolk sac, but a separate cavitation event cannot be excluded. By the CS6 stage, the SYS develops as a pocket of the primary yolk sac bounded by the visceral endoderm and part of the parietal endoderm. The SYS becomes encased by investing mesenchyme, known as the extraembryonic mesenchyme. The SYS expands to occupy the abembryonic space of the embryonic disc, displaces the primary yolk sac and contributes to the allantoic diverticulum in the body stalk connecting the embryo to the chorionic tissues (Figure 2).

There is considerable uncertainty about the origin of the extraembryonic mesenchyme. The timing of secondary yolk-sac formation is close to the first appearance of a thickened area at one end of the epiblast that will give rise to the primitive streak. Outside of this area there are clearly already migrating extraembryonic mesoderm cells that will associate with the visceral endoderm to form the SYS and with the trophoblast to form the chorionic villi (Figure 1A). The proximity of the mesenchyme to the yolk-sac endoderm, the morphological implication of cell trafficking from the endoderm to the mesenchyme (Enders and King, 1988) and the sharing of molecular (transcriptome) profile (Ross and Boroviak, 2020; Xiang et al., 2020) raise the possibility that the extraembryonic mesenchyme is derived from the hypoblast/endoderm of the yolk sac (Bianchi et al., 1993) (Figure 2). In rodents, extraembryonic mesoderm of the yolk sac and chorion are the earliest cells to leave the posterior region of the epiblast (Kinder et al., 1999), and lineage tracing shows no evidence of a contribution to these cells from the primitive endoderm (hypoblast) derivatives (Gardner and Rossant, 1979). The emergence of extraembryonic mesoderm from the primitive streak occurs concurrently with the recruitment of epiblast cells to the amniotic folds (Pereira et al., 2011), and the extraembryonic mesoderm contributes to the mesothelial tissue lining the amnion and the exocoelomic (yolk sac) cavity (Kinder et al., 1999). As the primate extraembryonic mesenchyme emerges prior to the appearance of the primitive streak and formation of the primary germ layers, it was presumed that the mesenchyme does not originate from the epiblast at gastrulation. However, the transcriptome analysis of peri-implantation embryos *in vitro* revealed that there may be focal dissemination of cells from the epiblast of cells that came to occupy the space near the visceral endoderm. These cells express markers of extraembryonic mesenchyme that was localized in between the yolk-sac endoderm and the trophoblasts (Xiang et al., 2020). In the *Cynomolgus* embryo at 10 dpf, single-cell analysis revealed the presence of mesoderm-like cells that are marked with signatures of the mouse extraembryonic mesoderm (Yang et al., 2021). Therefore, an early extravasation of extraembryonic mesenchyme from the embryonic disc in humans cannot be excluded (Luckett, 1978) and it might be coupled with the segregation of the amniotic epithelium from the post-implantation epiblast.

It is obviously impossible to resolve these issues by descriptive studies alone. Direct experimental lineage tracing and video recordings on human embryos cultured over this key period might be possible in future. Currently, however, although formation of both primary and secondary yolk sacs has been reported in cultured embryos (Xiang et al., 2020), the reproducibility and morphology of such structures is not great. Human stem-cell-derived models of hypoblast development and yolk-sac formation have not yet been fully developed. Authentic human hypoblast cell lines have yet to be extensively characterized although there have been two reports of culture conditions promoting either extraembryonic endoderm-like cells (Linneberg-Agerholm et al., 2019) or Yolk-sac endoderm-like cells (YSLCs) (Mackinlay et al., 2021) from human ES cells. During development, the SYS, besides serving essential embryotrophic functions for the conceptus prior to the establishment of functional placental support, also contributes to angiogenesis and hematopoiesis (Ross and Boroviak, 2020). A full understanding of the formation and function of the SYS is important for elucidating the initiation of the primitive hematopoiesis in the yolk-sac blood islands. While it has been possible to drive human ES cells toward the hematopoietic lineages by mimicking many of the signaling pathways worked out in the mouse, there are still challenges in generating transplantable human hematopoietic stem cells from ES cells, suggesting that a further investigation of the early events in the human yolk sac before the next phase of blood-cell development in the aorta-gonad-mesonephros of the embryo itself is warranted.

Recent approaches to the retrospective analysis of human developmental lineage relationships via accumulation of somatic mutations over time have been able to provide some clues as to the timing of lineage divergence during development. Several studies have shown evidence of asymmetric contributions of the progeny of the first two blastomeres to the later fetal and adult tissues, with ratios varying from 50:50 to 10:90 (Bizzotto et al., 2021; Fasching et al., 2021). This can be explained by asymmetric contributions of the early blastomeres to the ICM and TE lineages, which can also occur in the mouse embryo (Piotrowska et al., 2001). This approach can be extended to determining lineage trees and progenitor numbers in specific tissues. Whole genome sequencing of multiple single-cell-derived hematopoietic colonies from two human fetuses at 8 and 18 weeks, coupled with targeted sequencing of samples of other fetal tissues allowed lineage trees of embryonic and extraembryonic tissues to be developed (Chapman et al., 2021). Early separation of TE and ICM-derived lineages was confirmed, as predicted. A higher correlation of mutations in the primitive blood colonies to the extraembryonic mesoderm of the placental villi than to the tissues of the fetus itself is consistent with the origin of the primitive blood islands in the SYS outside of the fetus (Chapman et al., 2021). However, the authors claim that it supports a hypoblast origin for the blood islands. This is not really the case. The somatic lineage tree supports an early and separate origin of the extraembryonic mesoderm from the embryonic mesoderm but does not distinguish a hypoblast or epiblast origin for these cells. It would be unexpected for hematopoietic progenitors to derive from the yolk-sac endoderm in humans, especially since there is extensive work showing that epiblast-derived pluripotent stem cells (PSC) can differentiate into hematopoietic lineages (Lacaud et al., 2004).

Trophectoderm to chorionic villi

The earliest phases of the development of the human placenta are also still poorly understood. Comparative studies in mice have defined some of the candidate genes and pathways that are likely to be conserved, given the related hemochorial placentation in both species. However, there are major differences

between mice and humans in the development of the placenta from the very earliest phases of implantation in the uterus to the later complex villus tree. Indeed, placental morphogenesis is probably the most divergent process of organogenesis across all mammals, reflecting the relatively recent evolutionary origin of the placenta to support viviparity (Turco and Moffett, 2019). This diversity is also reflected in genomic differences in terms of divergent expansion of gene families involved in placental functions (Knox and Baker, 2008). Thus, understanding the human placenta requires studies in humans. Defects in placental development underlie many serious pregnancy problems including early embryo loss, preeclampsia, and intrauterine growth restriction. Poor placental function may also affect not just weight and maturity at birth but have long-term sequelae on predisposition to adult conditions such as hypertension and diabetes.

Descriptive studies from early human embryo material have provided the outline of the events of early human TE growth and differentiation (reviewed Knöfler et al. (2019); Turco and Moffett 2019). Initial attachment of the TE to the uterine epithelium takes place via the polar TE region overlying the ICM. These TE cells rapidly form an invasive syncytium—the primary syncytiotrophoblast (SCT)—which invades through the epithelium into the maternal endometrium, stimulating decidual formation which comes to surround the entire conceptus at around 14 days of development. There is still a layer of diploid trophoblast cells under the primary SCT which forms a cytotrophoblast shell around the developing embryo (Figure 1A). From this layer, projections of proliferating cells penetrate the SCT to form the primary villous cytotrophoblast (VCT), and the primary villous tree (Figure 2). Continued proliferation of the VCT drives formation of the anchoring villi in contact with the maternal decidua. Cytotrophoblast cell columns form at the distal tips of the anchoring villi, and extravillous trophoblast cells (EVT) begin to migrate into the maternal decidua to remodel the spiral arteries and enhance placental exchange. The chorionic villi of the mature placenta consist of a thin outer layer of SCT covering the VCT and containing a stromal core of fibroblasts, endothelial cells, and immune cells. The origin of the stromal core cells is likely extraembryonic mesoderm derived from early exiting cells from the primitive-streak anlagen. Suggestions of origin from the VCT directly by an epithelial to mesenchyme transition (EMT) or from the hypoblast (Bianchi et al., 1993) seem unlikely given comparisons with other species but cannot be disproven based on descriptive analysis alone.

The placenta at the stages from implantation on plays a major role in nutrition, gas exchange, and hormone production and as a potential source of signals directing embryo development. The lack of appropriate *in vitro* models of these early stages of placental development has hindered our understanding of this key organ. A recent comprehensive scRNA-seq analysis of the maternal-fetal interface during the first trimester has provided a great resource to begin to query the complexities of maternal-placental interactions (Vento-Tormo et al., 2018). However, valid experimental models to test mechanisms are not well established. Various cell lines, including choriocarcinoma-derived cells and primary cells from term placentae, have been useful in studying restricted aspects of placental development, such as syncytial formation or endocrine production (Turco and Moffett, 2019). However, none of the cell lines fully matches the *in vivo* situation. The recent development of the improved culture of intact human embryos over the implantation period (Deglincerti et al., 2016; Shahbazi et al., 2016; Xiang et al., 2020) and the isolation and characterization of cell lines and organoids from early human cytotrophoblast is changing the picture.

The outgrowth of human blastocysts over the implantation period *in vitro* has demonstrated the differentiation of the TE into the primary SCT and cytotrophoblast, but the length of the culture period would not be expected to be sufficient to model complex chorionic villus formation (Figures 1A and 1B). scRNA analysis of such cultures by supports identification of CT and primary SCT, with some evidence of EVT-like cells (Molè et al., 2021; Xiang et al., 2020), but there is little evidence of further differentiation and association with extraembryonic mesenchyme cells of any origin. More relevant models for studying post-implantation human trophoblasts have been the recently derived human TS cells and trophoblast organoids. Okae and colleagues first derived permanent cell lines from human blastocysts and from first-trimester placental villi that demonstrated many of the properties predicted for a TS cell, including expression of human-specific trophoblast markers and the ability to differentiate in culture into SCT and EVT under appropriate culture conditions (Okae et al., 2018).

Generation of 3D trophoblast organoids from first-trimester placentae has also recently been reported (Haider et al., 2018; Turco et al., 2018). The most extensively characterized structures were derived from EpCam-enriched cytotrophoblast cells and grown in Matrigel with an empirical mix of cytokines and inhibitors, with EGF being the most important. Complex villus-like structures were generated with both SCT and VCT layers (Turco et al., 2018). Switching cultures to Okae differentiation medium (Okae et al., 2018) allowed the outgrowth of EVT cells. Markers predicted to be specific to human villus cytotrophoblast were robustly expressed. TS cells and trophoblast organoids have strong potential to be able to model early placental development in both healthy and compromised pregnancies. TS cells offer the possibility of patient-specific cell lines from placental biopsies or CRISPR-generated mutant lines which could then be grown as 3D organoids. However, the villus structures of the organoids currently lack the stromal components of the villus core which would need to be added. It would also be important to be able to mimic the actual trophoblast-uterine stroma interactions to fully understand implantation and placentation. Recent generation of endometrial gland organoids (Luddi et al., 2020; Turco et al., 2017) offers an interesting potential model to explore in this regard.

Allocation of amniogenic cells and germline cells

During the development of the CS4–CS5-implanting blastocyst, the epiblast undergoes epithelialization and lumenogenesis to form an amniotic sac consisting of an amnionic epithelium of low cuboidal cells on the TE side of the amniotic cavity, which subsequently thins to a squamous epithelium and is invested by a mesothelial layer to form the amnion. On the hypoblast side, the epiblast cells form a pseudostratified columnar epithelium (the embryonic disc) (Figure 1A). Little is known of the molecular and cellular mechanism of the formation of the amniotic epithelium. It may be noted that formation of the amniotic epithelium in the ES-based human post-implantation amniotic-sac

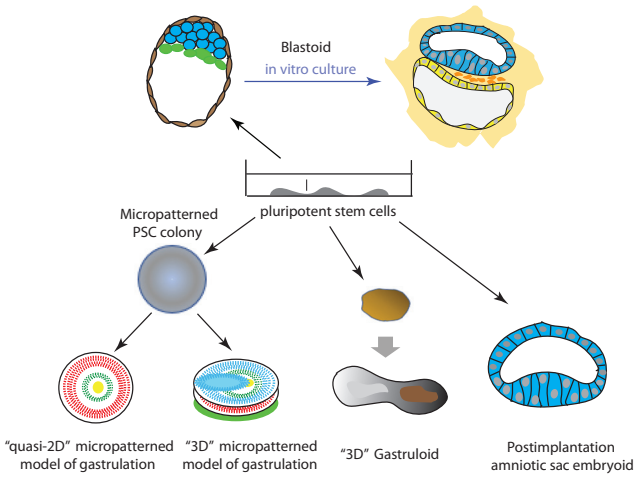


Figure 3. Experimental models for studying peri-implantation human development

Pluripotent stem cell (PSC)-based embryo models include the integrated models such as the blastoid, which on *in vitro* culture may generate a multitude of cell types and recapitulates the architecture of the peri-implantation human embryo, and the nonintegrated models, exemplified by the micropatterned PSC colony that develops into "quasi-2D" (pseudostratified/stratified cell layer) structures (Warmflash et al., 2014; Etoc et al., 2016; Phan-Everson et al., 2021) and "3D" (multiple cell layers that are reminiscent of germ layers) micropatterned models of gastrulation (Chhabra et al., 2019; Martyn et al., 2019), and the "3D" gastruloid and PASE (Matthews et al., 2021; Rossant and Tam, 2021).

embryoid (PASE) requires induction by BMP4 signal (Zheng et al., 2019), which may be sourced from extraembryonic tissue, such as the trophoblast and extraembryonic mesenchyme *in vivo*. It is not clear whether cells in the amniotic epithelium are allocated from a specific population of the epiblast and if there is an ongoing allocation of epiblast cells to the amniogenic population from the peripheral region of the embryonic disc (Figure 2). Lineage-tracing study in the mouse showed that the progenitors of the ectoderm cells of the amnion are localized in the proximal sector of the epiblast (Kinder et al., 1999), which is equivalent to the peripheral region of the human embryonic disc. The progenitors continue to contribute cells to the amniotic fold (Dobreva et al., 2018), which eventually forms a complete amnion across the proamniotic cavity (Pereira et al., 2011).

The early development of amnion in the human embryo is reputed to be coupled with the specification of the germline lineage prior to the onset of gastrulation. The presence of PGCs has been reported in a CS7 gastrula-stage human embryo (Tyser et al., 2021) and PGC-like cells were induced by BMP in the post-implantation amniotic sac embryoid (PASE) (Zheng et al., 2019). Studies in the *Cynomolgus* monkey embryo revealed that PGC-like cells are first detected in the presumptive posterior part of the amniotic epithelium (the nascent amnion), and, thereafter, among the mesenchyme in the vicinity of the posterior epiblast (Sasaki et al., 2016). Based on these findings, it was hypothesized that the human germline lineage may originate from the amnion epithelium and the posterior epiblast (Kobayashi and Surani, 2018). In the pregastrulation mouse embryo, progenitors of the PGCs are first detected in the proximal-posterior region of the epiblast (Saitou et al., 2005) among epiblast cells that are fated for amniotic ectoderm (Dobreva et al., 2018) and extraembryonic mesoderm (Kinder et al.,1999). Clonal analysis of the proximal epiblast cells of E6.2 pregastrulation embryos showed that single cells could contribute to multiple cell/tissue types including PGCs, amniotic epithelium, allantoic mesoderm, and posterior PS. However, there were no cases in which progenitors of amniotic ectoderm also contributed to PGCs (Dobreva et al., 2018). In the amniotic epithelium of the *Cynomolgus* embryo, two populations of amniotic cells are found: one population is marked by the expression of genes associated with epithelial development and differentiation, and the other displays high expression of pluripotency-related genes (Yang et al., 2021). This may suggest that the PGCs are specified in a subset of amniotic epithelial cells with multilineage potency and may not be derived secondarily from the amniogenic cell lineage. Going forward, it will be interesting to resolve the origin of the germline/PGCs in the human embryo and whether the multitude of signaling activities, such as BMP, WNT, and HIPPO, and the TFs that were inferred from the PASE model and PSCs (Kobayashi and Surani, 2018; Zheng et al., 2019; Saitou and Hayashi, 2021; Kojima et al., 2021; Kagiwada et al., 2021), are required for the induction of PGCs *in vivo*.

Signaling activity in germ layer differentiation and tissue patterning

Peri-implantation embryogenesis involves the transformation of the epiblast into the embryonic sac, and it heralds the onset of gastrulation, which drives the formation of the germ layers and the regionalization of germ-layer derivatives to establish the body plan of the developing embryo. Close interactions between the epiblast and extraembryonic cell types are key to this transition.

In the post-implantation mouse embryo, formation of the PS and initiation of germ layer formation are driven by signaling activities emanating from the extraembryonic tissues such as the extraembryonic ectoderm, which gives rise to chorionic ectoderm, and visceral endoderm and from within the epiblast. However, in primates, there is no clear equivalent to the TE-derived extraembryonic ectoderm of the mouse, which is a major source of BMP to induce gastrulation and primitive-streak formation. In humans and nonhuman primates, the amnion (amniotic epithelium) is reputed to behave like a signaling center of BMP activity to induce the differentiation of mesoderm in the *Cynomolgus* embryonic disc and the human PASE (Shao et al., 2017; Yang et al., 2021; Zheng et al., 2019).

In planar circular micropatterns derived from human pluripotent embryonic stem cells (hESC), which model the epiblast of the human embryo (Figure 3), the cellular response to BMP-signaling activity, mediated by localized modulation of BMP signaling by the antagonistic activity of Noggin, is associated with the induction of cells of ectoderm, mesoderm, and endoderm fates (Warmflash et al., 2014; Etoc et al., 2016; Phan-Everson et al., 2021). While cells with extraembryonic properties are induced in the periphery of the micropattern under different signaling conditions, it remains unclear if they are extraembryonic mesoderm, TE, cells or amnion-like cells (Chhabra et al., 2019; Chhabra and Warmflash, 2021). The induction of different germ-layer derivatives and their radial organization in the spatial order reminiscent of the rostral-caudal order of derivatives of the PS of the mouse (Tam and Loebel, 2007) showed that the

differentiation and patterning of germ-layer derivatives during embryogenesis may be influenced by the spatial availability of BMP signal (Etoc et al., 2016). It may be noted that the response to signaling activity in making cell-fate decision is influenced by the size of the cell population, as measured by cell number (Nemashkalo et al., 2017), suggesting that some form of community effect is in play to mediate the response of cells to signaling activity (Gurdon, 1988).

Modulating the level of BMP and WNT-signaling activity over time by chemical inhibition and varying the availability of signaling factors, concurrently with blocking TGFβ signaling, induced the differentiation of neuroepithelial cells, neural crest cells, placode, and surface ectoderm in the micropattern. These ectoderm cell types are also generated in the domain size and the spatial order reflective of the ectoderm cell populations in medial-lateral body axis *in vivo* (Britton et al., 2019). Modulating the level of BMP, WNT, and Nodal activity induced cells of all three germ-layer fates in the multilayered (≈3D) micropattern and elicited morphogenetic cell movements reminiscent of those in the PS (Chhabra et al., 2019; Martyn et al., 2019). Furthermore, supplementing activin to BMP and WNT in the micropattern generated organizer-like tissue, which could induce a secondary body axis containing cells of neural fates when transplanted into the host chick embryo (Martyn et al., 2018). Analysis of the development of PSC micropatterned models of gastrulation and the PASE (Figure 3) has shown that the signaling pathways driving gastrulation are likely to be conserved among the mammalian embryos. However, due to the absence of some specific embryonic and extraembryonic cell types in these experimental models (Posfai et al., 2021a), the source and geometry of signaling activity and cell interaction remain to be resolved. Future research may depend on the development of improved integrated embryo models, such as the blastoids that may be constituted with the full complement of embryonic and extraembryonic cells (Figure 3), so that the experimental findings could inform the developmental process of germ layer differentiation and tissue patterning during embryogenesis *in vivo*.

Embryonic tissue patterning during gastrulation

Gastrulation involves the allocation of germ-layer precursors via inductive histogenesis and morphogenetic cell movement at the PS and the patterning of the germ-layer tissues. In the mouse, gastrulation is governed by signaling activity from the visceral endoderm, the PS, and the organizer, in addition to that from the extraembryonic ectoderm, a tissue derived from the TE. Regionalized BMP activity in the mouse extraembryonic ectoderm is critical for regulating another signaling source, the visceral endoderm (Richardson et al., 2006). A specialized population, the anterior visceral endoderm (AVE), is a source of Nodal and WNT antagonists that promote anterior patterning and restrict mesoderm development to the posterior side (Hoshino et al., 2015; Yoshida et al., 2016). In the primate embryo, the amniotic epithelium (the likely equivalent of the proximal epiblast of the mouse) may act as the surrogate for the extraembryonic ectoderm, although a role for the cytotrophoblast population that is in close proximity to the embryonic disc cannot be excluded. Whether there is an equivalent of the mouse AVE in primate embryos is under active investigation. In *Cynomolgus* embryos, a marked thickening of the visceral endoderm (VE) at one side of the embryo has been noted and immunofluorescence studies of cultured embryos have indicated that the initial widespread expression of the Nodal inhibitor, CER1, in the VE becomes localized to a subset of VE cells (Ma et al., 2019; Yang et al., 2021). In the human peri-implantation embryo in culture, a subset of hypoblast cells expressing genes encoding the antagonists to WNT and Nodal/BMP signaling, are progressively regionalized to the prospective anterior region of the embryonic disc during development (Molè et al., 2021). Yolk-sac endoderm-like cells derived from human ES cell cultures were shown to express antagonistic factors against WNT and BMP signaling that could block the BMP-induced downregulation of human pluripotency factors in hESCs (Mackinlay et al., 2021), providing a possible future *in vitro* model for hypoblast-epiblast interactions in germ-layer formation.

The impact of signaling activity on tissue patterning in the embryonic disc has been evaluated in the PASE model. BMP4 and WNT3A induce cells with posterior PS characteristics whereas BMP4 and activin induce cells with anterior PS properties in the embryonic disc. Loss of ISL1-mediated BMP activity in the amniotic epithelium of the PESC led to failure to induce *T/BRA* expression in the epiblast (Yang et al., 2021) and blocking BMP and WNT activity anteriorized the epiblast population of the PASE (Zheng et al., 2019). These findings corroborate the patterning effect of BMP and WNT on the germ-layer precursors in the micropatterned model. However, neither experimental model is amenable for elucidating the impact of hypoblast-epiblast interaction on tissue patterning in the embryonic disc prior to onset of gastrulation or during gastrulation.

FROM GASTRULATION TO TISSUE PATTERNING IN THE BODY AXES

As gastrulation proceeds, the mouse embryo acquires patterning of the germ-layer tissues in the anterior-posterior, medial-lateral, and dorsal-ventral dimensions. The recent description and spatially resolved single-cell analysis of an intact gastrulation-stage human embryo (Tyser et al., 2021) indicates that there will be more insights into embryonic patterning to be gained from further direct analysis of human postgastrulation embryos. However, no experimental interventions in intact embryos beyond gastrulation are feasible or permissible. Thus, stem-cell-derived models of these stages of development, such as the 3D micropatterns and polarized gastruloids, gain in importance for recapitulating the developmental attributes of gastrulation and tissue patterning.

In contrast to the radial patterning of medial-lateral cell types of neuroepithelium, placodal progenitors, neural crest cells, and surface ectoderm in the micropatterned model (Britton et al., 2019), the imposition of mechanical constraint by conversion of the micropattern to a 3D tubular configuration led to patterning of the ectoderm derivatives into a neural-plate-like primordium and lateral ectoderm domains. The neural-plate-like primordium self-organized into the neural tube by elevation and fusion of the neural folds (Karzbrun et al., 2021). It is unclear, however, if the prior 3D patterning of the ectoderm tissues is critical for the morphogenesis of the neural tube. It

has been postulated that the mechanical force generated by the longitudinal growth of the neural plate may be instrumental for the closure of the neural folds during neurulation (Jacobson and Tam, 1982). Nevertheless, this study has shown that the 3D micropattern brings the stem cell model closer to the morphogenetic reality of the neural-plate primordium.

In the 3D hESC epithelial-cyst model, modulation of agonistic and antagonistic WNT-signaling activity following induction by BMP led to the polarization of SOX2+ and T/BRA+ cells in the cystic structure and the emergence of cell types with molecular signature of the PS and epithelial-mesenchymal transition (Simunovic et al., 2019). Similar spontaneous acquisition of SOX2-T polarity was observed in the 3D gastruloid model, derived from aggregates of PSCs treated with WNT agonists (Steventon et al., 2021). On further development, the polarized gastruloid forms an elongated embryo-body-like structure displaying spatially patterned gene expression signatures of posterior mesoderm lineages and early somitogenesis (Moris et al., 2020), suggesting the polarization of cell types in the gastruloid is developmentally upstream of embryonic patterning. The polarization of cell types in the mouse gastruloid has been taken as comparable with the initiation of anterior-posterior patterning of the body plan during mouse gastrulation. However, it is achieved without the cells traveling through the morphogenetic journey of PS-mediated gastrulation and experiencing the spatiotemporal signaling activity associated with germ layer formation. This has led to the suggestion that the acquisition of axis asymmetry may be uncoupled from any species-specific gastrulation activity and that axis development in the gastruloid reflects a conserved minimalist morphogenetic program of body-plan formation (Steventon et al., 2021). However, the emergence of the polarized *T*-expressing cell population in the human gastruloid might represent instead the generation of a multilineage progenitor-cell population with propensity toward axis growth and elongation similar to the postgastrulation posterior growth zone of the mouse embryo (Cambray and Wilson, 2002; Goldman et al., 2000; Tam and Tan, 1992). It is not clear that the rest of the cells in the gastruloid have acquired full anterior characteristics. It therefore remains questionable whether patterning cues acquired by cells in these gastruloids and the outcome of tissue patterning are fully developmentally relevant to the formation of the whole body plan in the early embryo (Rossant and Tam, 2021). Ideally, future investigation of embryonic patterning would be conducted in embryonic disc/gastrula models that integrate the spatial-temporal attributes of the putative signaling centers such as the amniotic epithelium, YSLCs, AVE-like cells, the PS and the organizer.

IMPERATIVES AND CHALLENGES TO THE ROAD AHEAD

Twenty years ago, there were no papers on human embryo development in *Developmental Cell* or any frontline journal. Human embryo research was highly restricted in many jurisdictions, including the use of embryos to generate stem cells. And yet it is vital to understand the processes of human development to improve IVF, prevent early embryo loss, and explore developmental diseases and pregnancy disorders. Although the basic blueprint of development as laid down in the mouse and nonhuman primates provides the starting point for understanding human development, there are differences, and they matter. Today we have more and better tools, such as the embryo models to study early human development and develop a multi-omic molecular database of normal development at fine levels of temporal, spatial, and lineage resolution. Descriptive and molecular studies on precious existing and newly collected embryo samples from these important stages may be forthcoming in the next few years. They will provide important baseline data to compare with the growing number of stem-cell-based embryo models of these stages of development. Recent effort in the construction of a Human Developmental Cell Atlas by single-cell molecular profiling will provide in-depth spatiotemporal information for building time-series trajectories of cells and tissues in the timeline from fertilization to birth (Haniffa et al., 2021). Data generated using experimental models of blastocyst and early post-implantation, gastrulation, and post-gastrulation embryos and from whole embryos at weeks 6–8 (CS17–20 stage in the late embryonic period) will enrich the knowledge of lineage phylogeny in peri-implantation- to post-gastrulation stage embryos. While experimental prospective lineage tracing is technically difficult in human embryos and embryo models, retrospective lineage tracing using mutational fingerprints in cells will provide additional insights into the lineage relationships of cells in tissues, organs, and body parts. The data reported already enlighten our understanding of the allocation of progenitors from the epiblast for germ layers, germ cells, extraembryonic tissues, and specific organs during gastrulation (Bizzotto et al., 2021; Coorens et al., 2021) and are relevant for the construction of lineage trajectories in peri-implantation development. With the development of multi-dimensional human biomolecular cell and tissue atlases that incorporate spatial attributes and multilayered omics data spanning genomics, transcriptomics, proteomics, epigenomics, to metabolomics (HuBMAP, 2019), it should be feasible in the future to construct a trans-regulatory network for lineage development that encompasses the allocation of progenitor cells to specific cell lineages and subsequent cell differentiation along the developmental trajectory of the lineages and tissue morphogenesis in the human embryo.

DECLARATION OF INTERESTS

J.R. and P.P.L.T. are members of the Advisory Board of *Developmental Cell*.

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